Paper : DSE-A-2

(Operation Research)

Full Marks : 50

The figures in the margin indicate full marks.

Answer ***question no.*** **1** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5
   (a) What is slack variable?
   (b) When is a problem said to be a degenerate problem?
   (c) What are the different elements of a two person zero sum game?
   (d) What cautions are taken when there are negative costs in an assignment problem?
   (e) Define basic feasible solution of a linear programming problem.
   (f) What do you mean by Max-Min value in game theory?
   (g) What are the corner points in a Simplex method?
   (h) Why are artificial variables needed in a Simplex problem?

2. Find the initial basic feasible solution for the following assignment problem : 10

Project

		I	II	III	IV
Person	1	4	5	3	2
	2	1	4	–2	3
	3	4	2	1	–5

3. (a) Define a transportation problem.
   (b) Find the minimum cost of transportation problem of the following : 4+6

Destination

		D_1	D_2	D_3	a_i
Origin	0_1	4	3	2	10
	0_2	1	5	0	13
	0_3	3	8	6	12
	b_j	8	5	4	

4. Solve the Following LPP : 10

Min, $Z = 4x_1 + 3x_2$

Subject to $3x_1 + 4x_2 \leq 28$

$-x_1 + 2x_2 \leq -21$

$x_1 + 2x_2 \geq 32$

$x_1, x_2 \geq 0.$

5. Solve Graphically : 10

Maximize, $Z = 3x_1 + 4x_2$

Subject to $x_1 - x_2 \geq 0$

$-x_1 + 3x_2 \leq 3$

and $x_1, x_2 \geq 0.$

6. Solve the following LPP using Simplex method : 10

Maximize, $Z = 6x_1 + 8x_2$

Subject to $30x_1 + 20x_2 \leq 300$

$5x_1 + 10x_2 \leq 110$

$x_1, x_2 \geq 0.$

7. (a) Briefly explain the steps to convert a primal problem of an LPP to its dual problem.

(b) Find the dual of the following problem : 4+6

Maximize, $Z = 2x_1 + 3x_2 + x_3$

Subject to $4x_1 + 3x_2 + x_3 = 6$

$x_1 + 2x_2 + 5x_3 = 4$

$x_1, x_2, x_3 \geq 0.$

8. Find the optimum strategies for A and B and the value of the game : 10

	B			
A	2	8	3	5
	1	4	8	9
	6	3	7	11